

The Linear Algebra Version of the Chain Rule

Ata Kabán

School of Computer Science
University of Birmingham



Outline

Linear Maps

Matrices

Linear Maps and Matrices

Jacobian and Linear Approximation

The Chain Rule

Examples

Definition

A **linear map** $F : V^n \rightarrow V^m$ satisfies:

$$F(c\mathbf{x}) = cF(\mathbf{x}), \quad F(\mathbf{x} + \mathbf{x}') = F(\mathbf{x}) + F(\mathbf{x}')$$

Component form:

$$F(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

Examples of Linear Maps

► $m = 1, n = 3 : y = ax_1 + bx_2 + cx_3$

► $m = 2, n = 3 : \begin{cases} y_1 = ax_1 + bx_2 + cx_3 \\ y_2 = dx_1 + ex_2 + fx_3 \end{cases}$

► $m = 3, n = 2 : \begin{cases} y_1 = ax_1 + bx_2 \\ y_2 = cx_1 + dx_2 \\ y_3 = ex_1 + fx_2 \end{cases}$

Composition of Linear Maps

If $F : V^k \rightarrow V^n$ and $G : V^n \rightarrow V^k$ are linear, then:

$$(F \circ G)(\mathbf{x}) = F(G(\mathbf{x}))$$

and $F \circ G$ is also linear.

Definition of a Matrix

An $m \times n$ matrix is a rectangular array:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

- ▶ m rows, n columns.
- ▶ Column vectors: $n \times 1$, Row vectors: $1 \times n$.

Matrix Multiplication

Given A of size $m \times k$ and B of size $k \times n$:

$$(AB)_{ij} = \sum_{s=1}^k a_{is}b_{sj}$$

- ▶ Dimensions: $(m \times k)(k \times n) = (m \times n)$
- ▶ Associative, generally non-commutative.

Q: have you ever thought about why?

Example of Matrix Multiplication

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 6 \\ 4 & -3 & 3 \end{pmatrix}$$

Connecting Linear Maps and Matrices

Treat vectors $\mathbf{x} \in \mathbb{R}^n$ as column vectors. Given an $m \times n$ matrix A , we associate it to the following linear map:

$$F_A(\mathbf{x}) = A\mathbf{x} = \mathbf{y}$$

That is, $y_i = \sum_{j=1}^n a_{ij}x_j$.

Composition and Matrix Product

If $F : V^k \rightarrow V^n$ is a linear map given by the matrix A and $G : V^n \rightarrow V^k$ is a linear map given by matrix B then the composition $F \circ G$ is given by the matrix AB .

In other words: If $F(\mathbf{y}) = A\mathbf{y}$ and $G(\mathbf{x}) = B\mathbf{x}$, then:

$$(F \circ G)(\mathbf{x}) = A(B\mathbf{x}) = (AB)\mathbf{x}$$

Thus the composition corresponds to the matrix product.

Jacobian Matrix

For a map $F = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)) : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian at $\mathbf{x}_0$ is:

$$J_F(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}_0) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}_0) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}_0) \end{pmatrix}$$

The Chain Rule Theorem (Matrix Form)

Given two differentiable maps $F : D \rightarrow \mathbb{R}^m$, in components $F = (f_1(y_1, \dots, y_k), \dots, f_m(y_1, \dots, y_k))$, and $G : E \rightarrow \mathbb{R}^k$, in components $G = (g_1(x_1, \dots, x_n), \dots, g_k(x_1, \dots, x_n))$, with $E \subset \mathbb{R}^n$ and $D \subset G(E) \subset \mathbb{R}^k$, then we have:

$$J_{F \circ G} = J_F J_G$$

The Chain Rule Theorem (Matrix Form)

Given two differentiable maps $F : D \rightarrow \mathbb{R}^m$, in components $F = (f_1(y_1, \dots, y_k), \dots, f_m(y_1, \dots, y_k))$, and $G : E \rightarrow \mathbb{R}^k$, in components $G = (g_1(x_1, \dots, x_n), \dots, g_k(x_1, \dots, x_n))$, with $E \subset \mathbb{R}^n$ and $D \subset G(E) \subset \mathbb{R}^k$, then we have:

$$J_{F \circ G} = J_F J_G$$

Note: F and G need not be linear!

Proof Sketch

$$(J_{F \circ G})_{ij} = \frac{\partial f_i(g_1(x), \dots, g_k(x))}{\partial x_j} = \sum_{s=1}^k \frac{\partial f_i}{\partial y_s} \frac{\partial y_s}{\partial x_j}$$

which is the ij -th entry of $J_F J_G$.

Example 1

$$z = f(x, y, z) = f(\mathbf{x}), \quad \mathbf{x} = \mathbf{r}(t)$$

$$J_{f \circ \mathbf{r}}(x(t_0), y(t_0), z(t_0)) = \nabla f(x(t_0), y(t_0), z(t_0)) \begin{pmatrix} x'(t) \\ y'(t) \\ z'(t) \end{pmatrix}$$

$$\begin{aligned} &= f_x(x(t_0), y(t_0), z(t_0))x'(t_0) + f_y(x(t_0), y(t_0), z(t_0))y'(t_0) \\ &\quad + f_z(x(t_0), y(t_0), z(t_0))z'(t_0) \end{aligned}$$

Example 2

$$z = f(x, y), \quad x = g(s, t), \quad y = h(s, t) :$$

$$J_{f \circ (g, h)} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial g}{\partial s} & \frac{\partial g}{\partial t} \\ \frac{\partial h}{\partial s} & \frac{\partial h}{\partial t} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} & \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t} \end{pmatrix}$$

Example 3

$$z = f(x_1, \dots, x_n), \quad x_1 = g_1(t_1, \dots, t_m), \quad x_2 = g_2(t_1, \dots, t_m), \dots, \\ x_n = g_n(t_1, \dots, t_m)$$

We set $G = (g_1, \dots, g_n)$ and obtain:

$$J_{f \circ G} = \nabla f J_G = \left(\frac{\partial f}{\partial x_1} \frac{\partial f}{\partial x_2} \cdots \frac{\partial f}{\partial x_n} \right) \begin{pmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} & \cdots & \frac{\partial x_1}{\partial t_n} \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} & \cdots & \frac{\partial x_2}{\partial t_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_m}{\partial t_1} & \frac{\partial x_m}{\partial t_2} & \cdots & \frac{\partial x_m}{\partial t_n} \end{pmatrix} \\ = \begin{pmatrix} \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_1} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_1} + \cdots + \frac{\partial f}{\partial x_n} \frac{\partial x_n}{\partial t_1} \\ \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_2} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_2} + \cdots + \frac{\partial f}{\partial x_n} \frac{\partial x_n}{\partial t_2} \\ \vdots \\ \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_m} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_m} + \cdots + \frac{\partial f}{\partial x_n} \frac{\partial x_n}{\partial t_m} \end{pmatrix}$$

Summary

- ▶ Differentiable maps are locally linear.
- ▶ Matrix algebra provides the language for linearization.
- ▶ The chain rule expresses composition of derivatives as matrix multiplication.

Videos (optional):

- ▶ Multi-variable chain rule:
https://www.youtube.com/watch?v=_H30M201qFQ
- ▶ Detailed proof of multi-variable chain rule:
<https://www.youtube.com/watch?v=mnPqoTm0V4w>